SCHOOL OF ENGINEERING

Dayananda Sagar University Bengaluru

Prediction and Severity Classification of Wilson's Disease using Machine Learning

Minor Project | Dept. of CSE (AI&ML)

Students Details:

Shreyas D C (ENG23AM0071)

Sneha B (ENG23AM0073)

Poorvi C Shetty(ENG23AM00260)

Mentored By:

Upasana Shil

Assistant professor

CSE(AIML)

Objectives

- To develop a machine learning model for early prediction of Wilson's Disease using biochemical liver function test parameters.
- To classify the severity of Wilson's Disease into mild, moderate, and severe stages using supervised classification algorithms.
- To compare the performance of multiple ML algorithms (Random Forest, SVM, XGBoost, Logistic Regression) and identify the best-performing model.

Introduction

- Wilson's Disease is a rare genetic disorder (autosomal recessive) caused by mutations in the ATP7B gene, leading to abnormal copper accumulation in the liver, brain, and other vital organs.
- Problem Statement: Current diagnosis of Wilson's Disease relies on invasive liver biopsy and complex scoring systems (Leipzig Score), making early detection difficult, especially in resource-limited settings.
- Available Solutions: Blood tests (serum ceruloplasmin, 24-hour urine copper), genetic testing, MRI brain scans, and liver function tests are used, but none individually provide sufficient accuracy.
- Gaps in Current Solutions: Existing diagnostic approaches lack automation, are expensive, time-consuming, and cannot effectively classify disease severity. Machine Learning can bridge this gap by utilizing routine LFT parameters.
- This project proposes an ML-based system to predict Wilson's Disease and classify its severity (mild/moderate/severe) using the Indian Liver Patient Dataset, enabling faster and cost-effective diagnosis.

Literature Review

S.No.	Details of Paper	Objectives, Dataset used, Methodology/Models used	Results/Advantages/Limitations
1	Ala, A. et al., 2007. Wilson's disease. The Lancet, 369(9559), pp.397-408.	To comprehensively review pathophysiology, diagnosis, and treatment of Wilson's Disease using clinical and laboratory parameters including ceruloplasmin, 24-hr urine copper, and liver biopsy.	Leipzig Scoring system established as standard diagnostic criterion. Early treatment with D-penicillamine reduces mortality. Liver transplantation curative for end-stage disease. Limitation: No automated ML approach.
2	Ramana, B.V. et al., 2011. A critical study of selected classification algorithms for liver disease diagnosis. IJDMS, 3(2).	To apply and compare Naive Bayes, SVM, and Decision Tree classifiers on the Indian Liver Patient Dataset for early liver disease detection using 10 biochemical features.	Naive Bayes achieved 71% accuracy. SVM outperformed with 75% accuracy. Dataset class imbalance identified as key limitation. Provided baseline benchmarks for liver disease ML classification.

Literature Review

S.No.	Details of Paper	Objectives, Dataset used, Methodology/Models used	Results/Advantages/Limitations
3	Bendi, V.S.S. and Murthy, C.S., 2018. A machine learning approach to predict liver disease. IJSRP, 8(5).	To evaluate Random Forest and Logistic Regression on ILPD for liver disease prediction. Dataset: 583 patients. Features: ALT, AST, Bilirubin, Albumin, Alkaline Phosphotase.	Random Forest achieved 85% accuracy and outperformed Logistic Regression (80%). Highlighted the significance of AST/ALT ratio. Limitation: No severity classification attempted.
4	Gunasundari, S. and Janakiraman, S., 2016. Comparison and performance evaluation of Naive Bayes and Logistic Regression. ICTACT JCSC, 6(3).	To compare probabilistic and regression-based classifiers for liver disease prediction using clinical parameters. Dataset used was ILPD with 10 features and 583 instances.	Logistic Regression showed superior precision (82%) while Naive Bayes had higher recall (79%). Concluded that model selection depends on clinical priority. No cross-validation performed.

Literature Review

S.No.	Details of Paper	Objectives, Dataset used, Methodology/Models used	Results/Advantages/Limitations
5	Breiman, L., 2001. Random Forests. Machine Learning, 45(1), pp.5-32.	To introduce Random Forest as an ensemble method combining multiple decision trees using bagging and random feature selection for classification and regression tasks.	Random Forest demonstrated superior accuracy over single decision trees and reduced overfitting. Proved robust to noise and missing values. Foundation algorithm for this project's primary classifier.
6	Chen, T. and Guestrin, C., 2016. XGBoost: A scalable tree boosting system. KDD 2016, ACM, pp.785-794.	To propose XGBoost, a gradient boosting framework with regularization for scalable and accurate supervised classification. Evaluated on multiple benchmark datasets including medical datasets.	XGBoost achieved state-of-the-art performance on 10 of 29 benchmark datasets. Regularization prevented overfitting. Highly suitable for tabular medical data. Used as secondary classifier in this project.

Literature Review

S.No.	Details of Paper	Objectives, Dataset used, Methodology/Models used	Results/Advantages/Limitations
7	Nahar, N. and Ara, F., 2018. Liver disease prediction using decision tree techniques. IJDMS, 10(1).	To apply C4.5, CART, and Random Tree decision tree algorithms for liver disease prediction on ILPD. Analyzed impact of pruning on accuracy and evaluated feature importance rankings.	C4.5 achieved 79.2% accuracy, best among decision tree variants. Total Bilirubin and ALT identified as top features. Limitation: Decision trees prone to overfitting without ensemble methods.
8	Giannini, E.G., Testa, R. and Savarino, V., 2005. Liver enzyme alteration: a guide for clinicians. Canadian Medical Association Journal, 172(3).	To analyze the diagnostic significance of ALT/AST ratio as a liver disease indicator. Dataset comprised clinical liver function test records from hospital patients with various hepatic conditions.	ALT/AST ratio below 1.0 strongly indicates liver fibrosis. High diagnostic value confirmed. Limitation: Limited to specific patient populations. Directly justifies the ALT_AST_Ratio engineered feature in Module 3.

Literature Review

S.No.	Details of Paper	Objectives, Dataset used, Methodology/Models used	Results/Advantages/Limitations
9	Gao, J., Zhang, Q. and Li, W., 2020. Liver disease severity classification using deep learning on clinical records. Computers in Biology and Medicine, 125.	To propose a multi-class liver disease severity classification system using enzyme level thresholds on clinical liver patient data. Classified patients into Mild, Moderate and Severe categories.	Achieved 84% accuracy in severity classification. Bilirubin and enzyme levels confirmed as best severity indicators. Limitation: Deep learning model requires large data. Conceptually identical to Module 5 severity classification in our project.
10	Lichman, M., 2013. UCI Machine Learning Repository. University of California, Irvine. Available at: http://archive.ics.uci.edu/ml.	To provide a publicly accessible repository of machine learning datasets for researchers. The Indian Liver Patient Dataset (ILPD) is hosted here with 583 instances, 10 features, and binary classification labels.	ILPD widely used as benchmark for liver disease ML research. Dataset limitations: 71.5% class imbalance, no copper or ceruloplasmin values. This project uses ILPD as proxy dataset for Wilson's Disease.

Dataset

- Dataset: Indian Liver Patient Dataset (ILPD) from UCI Machine Learning Repository, containing 583 patient records collected from Andhra Pradesh, India.
- Features (10): Age, Gender, Total Bilirubin, Direct Bilirubin, Alkaline Phosphotase, Alamine Aminotransferase (ALT), Aspartate Aminotransferase (AST), Total Proteins, Albumin, Albumin-Globulin Ratio.
- Target Variable: Binary label - 1 (Liver Patient / Wilson's Disease positive) and 2 (Non-patient). 416 positive cases and 167 negative cases.
- Key Parameters Relevant to Wilson's Disease: Elevated ALT, AST, Alkaline Phosphotase indicate hepatic copper toxicity. Low Albumin and A/G Ratio reflect liver synthetic dysfunction. Raised Bilirubin indicates cholestasis.
- Sample Data: Age=65, Gender=F, TBil=0.7, DBil=0.1, ALP=187, ALT=16, AST=18, TP=6.8, Alb=3.3, A/G=0.9 → Patient; Age=62, Gender=M, TBil=10.9, DBil=5.5, ALP=699, ALT=64, AST=100 → Patient.

Methodology

- COMPLETED - Step 1 (Data Collection): Indian Liver Patient Dataset (ILPD) loaded successfully - 583 records, 10 features, binary target variable (1=Patient, 2=Non-Patient).
- COMPLETED - Step 2 (Exploratory Data Analysis): Class distribution analyzed (416 positive, 167 negative). Gender distribution, feature ranges, and outliers identified using boxplots and histograms.
- COMPLETED - Step 3 (Data Preprocessing): Missing values handled (4 null values in Albumin_Globulin_Ratio filled with median). Gender encoded (Male=1, Female=0). Features normalized using MinMaxScaler.
- PLANNED - Step 4 (Feature Selection): SelectKBest (chi-squared) and correlation matrix to identify top predictive features. Expected key features: ALT, AST, Alkaline Phosphotase, Total Bilirubin, Albumin.
- PLANNED - Step 5 (Model Training & Evaluation): Train Random Forest, SVM, XGBoost, and Logistic Regression with 80/20 split and 5-fold cross-validation. Evaluate using Accuracy, F1-Score, and ROC-AUC.

Results

- NOTE: Model training and evaluation are yet to be completed. The following are expected/assumed results based on literature review and similar studies on the ILPD dataset.
- Expected: Random Forest is anticipated to achieve the highest accuracy (~88-92%) due to its robustness to class imbalance and ability to handle correlated features like ALT and AST.
- Expected: XGBoost is expected to perform comparably (~87-90% accuracy) with high precision. SVM and Logistic Regression are expected to achieve ~82-86% accuracy.
- Expected Top Features (based on domain knowledge): Alkaline Phosphotase, AST, ALT, Total Bilirubin, and Albumin are anticipated to be the strongest predictors based on Wilson's Disease pathophysiology.
- Severity Classification (Planned): Mild (AST/ALT < 1.5, Bilirubin < 2 mg/dL), Moderate (Bilirubin 2-5 mg/dL), Severe (Bilirubin > 5 mg/dL or Albumin < 2.5 g/dL) - labels to be assigned post-training.

Conclusions

- Expected Advantage: The proposed ML-based system is anticipated to predict Wilson's Disease with over 88% accuracy using non-invasive routine liver function test parameters, offering a faster and low-cost alternative to invasive liver biopsy and Leipzig Scoring.
- Current Limitation: Work is in progress; model training and severity classification are yet to be implemented. The ILPD dataset is a proxy for Wilson's Disease and lacks disease-specific markers such as serum ceruloplasmin and 24-hr urine copper values.
- Future Scope: Upon project completion, the system will be extended by integrating genetic markers (ATP7B mutation), deep learning models, and a web-based diagnostic interface for real-time clinical decision support.

References

- 1. Ala, A., Walker, A.P., Ashkan, K., Dooley, J.S. and Schilsky, M.L., 2007. Wilson's disease. The Lancet, 369(9559), pp.397-408.
- 2. Hasan, M.K. et al., 2020. Diabetes prediction using ensembling of different ML classifiers. IEEE Access, 8, pp.76516-76531.
- 3. Bendi, V.S.S. and Murthy, C.S., 2018. A machine learning approach to predict liver disease. IJSRP, 8(5).
- 4. Ramana, B.V., Babu, M.S.P. and Venkateswarlu, N.B., 2011. A critical study of selected classification algorithms for liver disease diagnosis. IJDMS, 3(2).
- 5. Singh, A. and Kumar, R., 2020. Heart disease prediction using machine learning algorithms. ICCCIS 2020, IEEE.
- 6. Nahar, N. and Ara, F., 2018. Liver disease prediction by using different decision tree techniques. IJDMS, 10(1).
- 7. Lichman, M., 2013. UCI Machine Learning Repository. Irvine, CA: University of California.
- 8. Gunasundari, S. and Janakiraman, S., 2016. Comparison and performance evaluation of Naive Bayes and Logistic Regression classifiers. ICTACT JCSC, 6(3).
- 9. Breiman, L., 2001. Random Forests. Machine Learning, 45(1), pp.5-32.
- 10. Chen, T. and Guestrin, C., 2016. XGBoost: A scalable tree boosting system. KDD 2016, ACM, pp.785-794.

Thank You